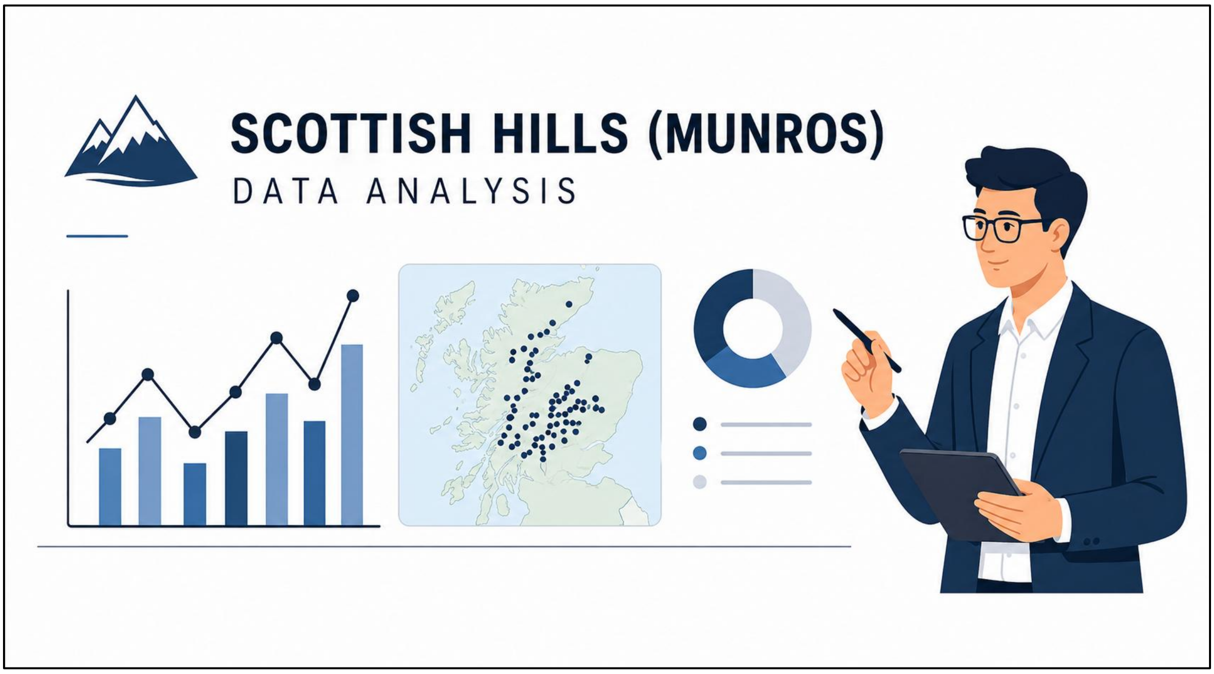

Data Analytics Project: Scottish Hills Analysis

Submitted By: TEAM 1

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Dataset Used: scottish_hills.csv
This document outlines a structured, end-to-end data analysis project based on height and location of Scottish mountains from the Scottish hills dataset. The project is divided into five scenarios, increasing in difficulty from basic data handling to advanced analysis and visualization, ensuring comprehensive skill coverage.

Project Goal: To analyze Scotland hills with their geographic information, and uncover insights about the distribution, elevation range of hills using the `Scottish_hills` dataset.

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts)

🎯 Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling (Mean/Mode), Data Cleaning, Type Conversion.
Pandas Operations	Column Selection, Filtering, Grouping, Aggregation, Value Count, Feature Engineering (Height Category creation)
NumPy Calculations	Array conversion, Numerical analysis using (np.diff) for height differences
Visualization	Trend Analysis, Line Chart, Bar Chart, Pie Chart, Histogram, Stacked Bar Chart, Saving Plots

Scenario 1: Data Loading & Cleaning (🟢 Easy)

Description:
This scenario involves the systematic loading and preprocessing of the dataset using the Pandas library. The dataset is imported from a CSV file, and an initial inspection is performed by displaying the first few records and column names. Data quality is assessed by identifying missing values in critical columns such as *Height* and *Region*. Appropriate data cleaning techniques are applied, including type conversion of the Height column to ensure numerical consistency. Missing values are then handled using statistical measures such as mean and mode to maintain data integrity.

Task	Description
1.	Load the `scottish_hills` dataset using Pandas.
2.	Display the first 5 rows (`head()`) of the dataset.
3.	Check for missing values in `Height`, and `Region`.
4.	Handling missing values: Fill the missing values `Height` – Use mean, `Region` – Use mode
5.	Convert `Height` column to numeric if required.

Results for Scenario 1:

```
First 5 Rows:
                    Hill Name  Height  ...    Osgrid      Region
0      A' Bhuidheanach Bheag   936.0  ...  NN660775  South-East
1              A' Chailleach   997.0  ...  NH136714  North-West
2              A' Chailleach   929.2  ...  NH681041  North-East
3  A' Chraileag (A' Chralaig)  1120.0  ...  NH094147  North-West
4            A' Ghlas-bheinn   918.0  ...  NH008231  North-West

[5 rows x 6 columns]
```

```
Missing Values After Cleaning:
Hill Name     0
Height        0
Latitude      0
Longitude     0
Osgrid        0
Region        0
dtype: int64
```

Conclusion:

The dataset has been successfully pre -processed and prepared for further analytical tasks. Missing values were effectively addressed, resulting in a clean and reliable dataset. The introduction of the Region column enhances the dataset's structure by enabling geographical categorization. These preprocessing steps significantly improve the overall quality and usability of the data. Consequently, the dataset is now well-structured and suitable for advanced analysis and visualization.

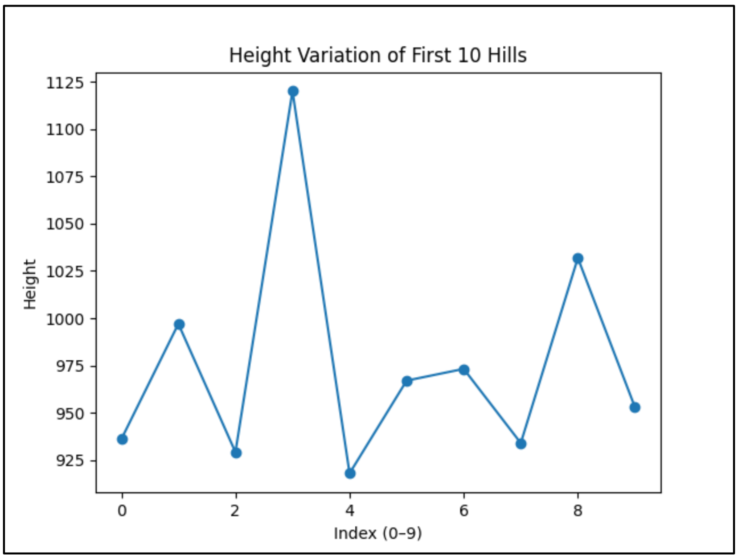
Scenario 2: Line Graph (Score Trend) + Save (▨ Easy)

Description:

In this scenario, line graph is created to visualize how the heights of hills vary. The dataset is first filtered to include only two columns: Hill Name and Height, and then limited to the first 10 records for simplicity. he Height column is converted into a NumPy array to enable numerical operations and plotting. A line graph is plotted.

Task	Description
1.	Select `Hill Name` and `Height`
2.	Take first 10 rows only.
3.	Convert the `Height` into a NumPy Array
4.	Plot a line graph with `X-axis – (index [0-9]), Y-axis – (Height)`
5.	Add the title and labels
6.	Save the graph: `plt.savefig("hill_heights_line.png")`.

Graph:



Conclusion:

The line graph provides a clear visual representation of how hill heights change across the first 10 entries in the dataset. It helps in quickly identifying:

- Variations in elevation
- Any increasing or decreasing trends
- Peaks (highest hills) and dips (lowest hills)

Although the graph is limited to only 10 hills, it effectively demonstrates how visualization can simplify data interpretation.

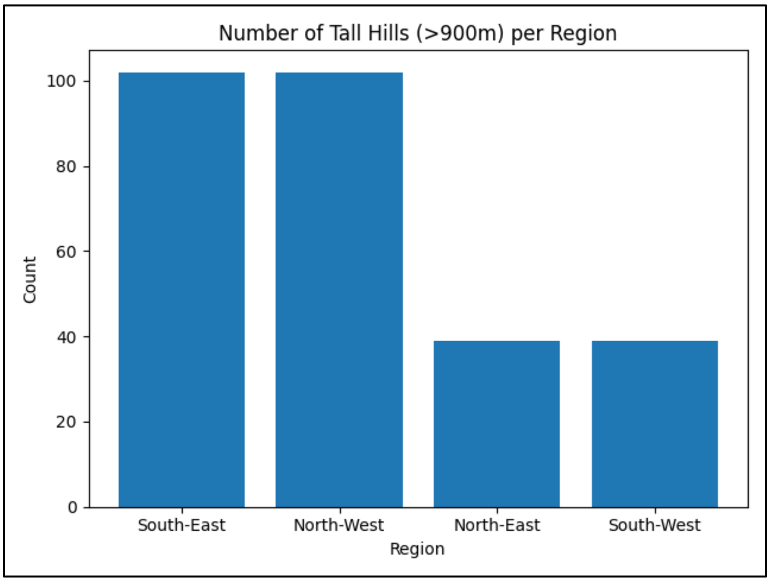
Scenario 3: **Filtering + Bar Chart + Save (▨ Medium)**

Description:

Here we will first filter the dataset to include only hills with a height greater than 900 meters. Then, we group the data by region and count how many tall hills are present in each region. we select the top regions with the highest counts. The results are converted into NumPy arrays for easier plotting. A bar chart is created where the X-axis shows the regions and the Y-axis shows the number of tall hills, and the labels are rotated for better readability.

Task	Description
1.	Filter the hills where `Height > 900`.
2.	Count the number of hills per `Region`.
3.	Select the top `Regions` using Pandas.
4.	Convert the results into NumPy arrays.
5.	Plot a bar chart with `Region` on the X-axis and count on the Y-axis.
6.	Rotate x-axis labels for readability and clarity.
7.	Save the graph: `plt.savefig("tall_hills_bar.png")`.

Graph:



Conclusion:

From this pie chart, we can conclude that some regions have higher number of tall hills compared to others. This shows that tall hills are not evenly distributed and are more concentrated in certain regions.

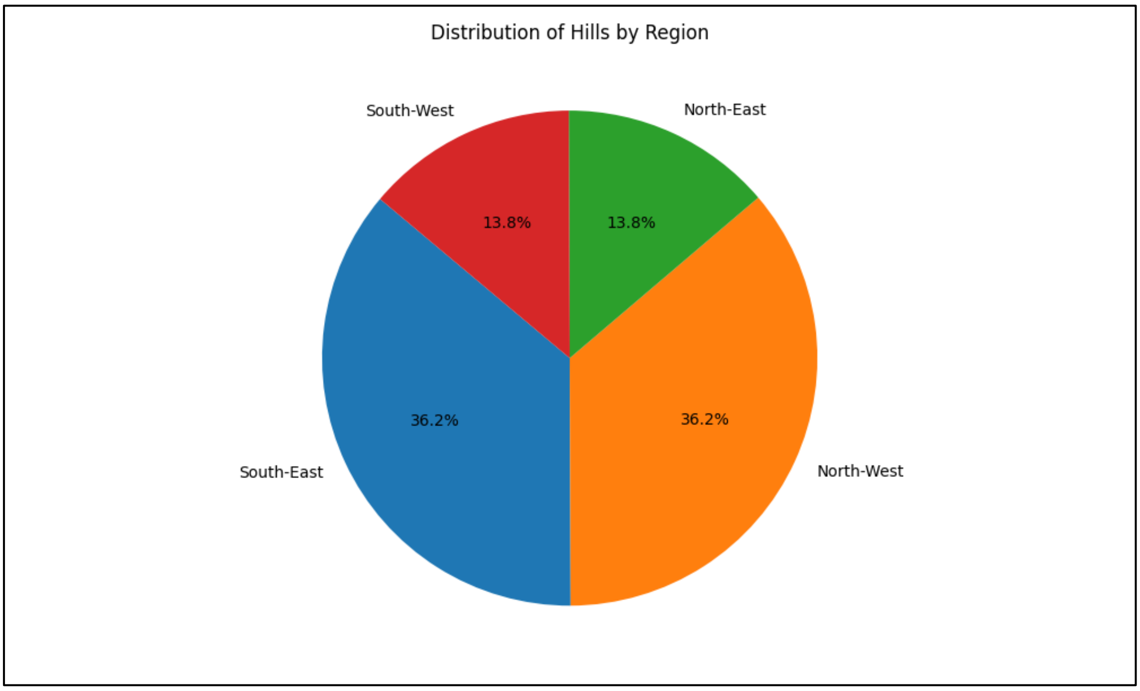
Scenario 4: Pie Chart (Region Distribution) + Save (🟡 Medium)

Description:

In this scenario, I have counted number of hills in each region using value_counts(). Then selected the top 5 regions and plotted them on a pie chart to show percentage distribution.

Task	Description
1.	Count the total number of hills per `Region`.
2.	Select the top 5 regions using Pandas.
3.	Prepare labels and values.
4.	Plot a pie chart.
5.	Add percentage labels using `autopct` (e.g., '%.1f%%').
6.	Save the graph: `plt.savefig("region_distribution.png")`.

Graph:



Conclusion:

The pie illustrates the distribution of hills across different regions, highlighting how hill counts vary geographically. It is evident that a few regions contribute a larger share of the total hills, while others have comparatively fewer. The top 5 regions together form the majority portion indicating that hill concentration is not evenly distributed. The distribution shows a slightly uneven pattern, with certain regions having a higher density of hills. Such insights are useful for geographical analysis and understanding regional terrain characteristics.

Scenario 5: Advanced Analysis + Multiple Graphs (● Hard)

Description:

Here in this scenario, we have performed advanced analysis on Scottish hills dataset by creating new feature called Height Category based on elevation ranges. We used NumPy to calculate differences between consecutive hill heights to understand variation patterns. Visuals were generated to study height of the hills across regions. All graphs were been saved for further analysis. Key insights such as tallest region and most common height category were identified.

Task	Description
1.	Create a new column `Height Category: Height >= 1000 (Very High), 800 – 900 (High), <800(Moderate)`
2.	Convert the `Height` column to NumPy Array. Calculate Height differences using `np.diff()`
3.	Plot a **Line Graph** for height trend of all hills.
4.	Plot a **Stacked Bar Chart** showing the count of `Height_category` per `Region`.
6.	Plot a **Histogram** showing the distribution of `Height`.
7.	Save the graphs: `plt.savefig("height_trend.png")`, `plt.savefig("height_category_stacked.png")`, `plt.savefig("height_histogram.png")`.
8.	**Identify**: Which region has tallest hills, Which category is most common, and Distribution pattern of heights.

Graph:

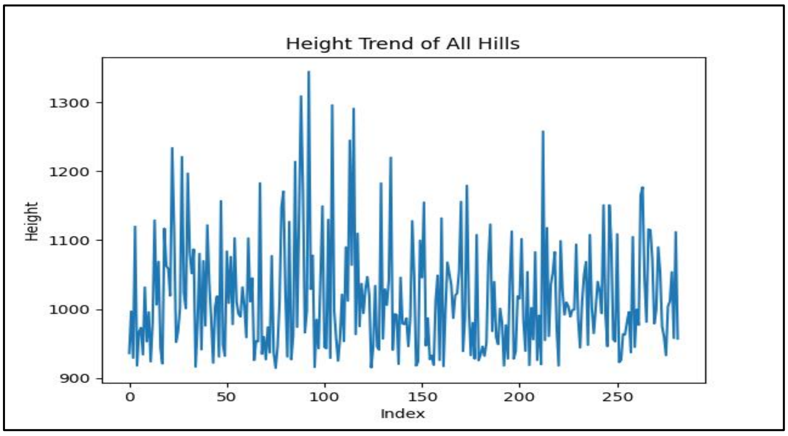


Fig 5.1 Height Trend of All Hills

```
Insights We have Got are:
Tallest Region (avg height): North-East
Most Common Height Category: High
```

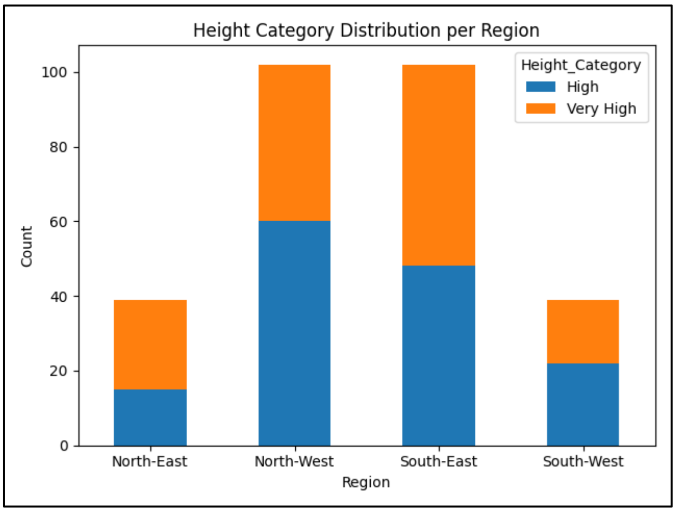


Fig 5.2 Height Category Distribution per Region

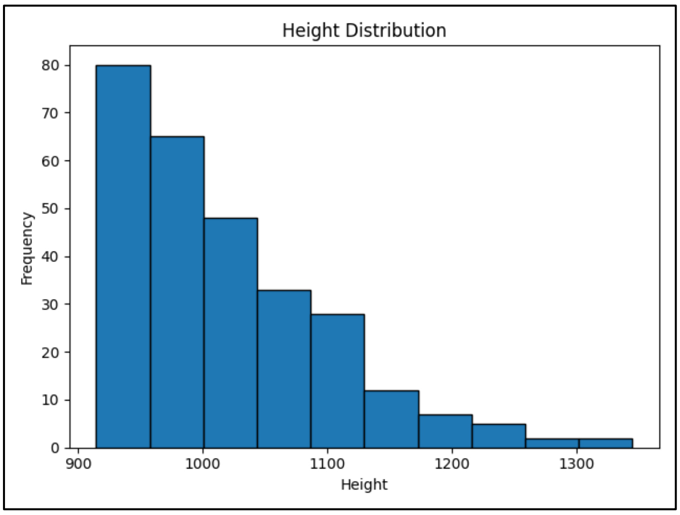


Fig 5.3 Height Distribution of Hills

Conclusion:

- The line graph shows that hill heights vary significantly across the dataset, indicating no uniform pattern but noticeable fluctuations in elevation.
- The stacked bar chart highlights how different regions contain varying proportions of height categories, with some regions having a higher concentration of very high hills.
- The histogram reveals that most hills fall within a specific height range, showing a clustered distribution rather than a uniform spread. From the analysis, it is evident that moderately high hills are more common compared to extremely tall ones.
- The tallest region, based on average height, stands out clearly indicating geographical elevation differences. The use of height categories helps simplify complex numerical data into meaningful groups for comparison,

Overall, the dataset shows a balanced but slightly skewed distribution towards mid-range elevations. These insights help in understanding regional elevation patterns and the overall structure of hill height in Scotland.

Final Conclusion:

This project successfully analyzes the Scottish Hills dataset using data preprocessing, numerical analysis, and visualization techniques. Initially, the dataset was cleaned by handling missing values, converting data types, and creating a new Region feature for better categorization. The line graph showed clear variations in hill heights, indicating fluctuations in elevation across different hills. The bar chart highlighted the distribution of tall hills across regions, helping identify areas with higher elevation concentration.

Further analysis using height categories revealed that medium-height hills contribute the most to the overall elevation. The histogram confirmed that most hills are clustered within a specific height range rather than being evenly distributed. Advanced analysis using NumPy helped in understanding variation patterns between consecutive hill heights.

Overall, the dataset shows a slightly skewed distribution toward mid-range elevations. The project demonstrates how combining preprocessing, feature engineering, and visualization can transform raw data into meaningful insights. It also highlights the importance of data analysis in understanding geographical patterns and trends effectively.

What you have learned

- I have learnt how to preprocess that data using pandas, including handling missing values and converting data types for consistency.
- We gained experience in feature engineering, such as creating the Region column and Height Categories, which made the dataset more meaningful and easier to analyze.
- We learned how to use NumPy for numerical operations, such as converting data into arrays and analyzing variations in hill heights.
- We developed skills in data visualization using Matplotlib, including line graphs, bar charts, pie charts, and histograms to represent different insights clearly.
- Finally, we gained an understanding of how to combine multiple techniques to complete an end-to-end data analytics project.